Prince **Asumadu**

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Research Interests

- Scientific Computing
- Numerical Linear Algebra
- Communication-Avoiding Algorithms
- Parallel & High-Performance Computing
- Krylov Subspace & Eigenproblem Methods
- Graph Algorithms & Analysis

Education

MPhil **Scientific Computing and Industrial Modelling** — Jan 2025 – Dec 2026 (expected)
National Institute for Mathematical Sciences (NIMS-KNUST), Ghana
- **Supervisor:** Dr. Richard Owusu

BSc **Mathematics** — Sep 2019 – Nov 2023
Kwame Nkrumah University of Science and Technology (KNUST), Ghana

Experience

Research

A Communication-Avoiding Parallel Spectral Clustering Pipeline, Jan 2026 – Present
National Institute for Mathematical Sciences, ***THESIS***

- Developing a spectral clustering pipeline for graph Laplacian with a communication-avoiding Krylov method, CA-Lanczos kernel to reduce synchronisations and global reductions.
- Implemented the classical Lanczos kernel in C, with eigenvalue accuracy checked against scipy eigsh symmetric eigensolver routine before extending to the s-step communication-avoiding formulation.

Research Assistant — Jan 2024 – Aug 2024
National Institute for Mathematical Sciences — Kumasi

- Ran computational experiments on the institute's cluster in support of ongoing research projects, handling job submission and the collection of results.
- Tested code for ongoing research projects at the institute, checking implementations and reporting issues to the researchers responsible.

Projects

Optimal Task Allocation using Modified Primal Simplex for KNUST Cleaning Staff Jan 2024 – May 2024
National Institute for Mathematical Sciences Project

- Formulated the allocation of tasks to employees as a linear programming problem, then implemented a modified primal simplex method to solve the resulting model.
- Compared the modified implementation against the classical primal simplex and the Hungarian algorithm on solution accuracy and runtime, and used the resulting allocation to produce a working duty roster for the cleaning staff.

Dynamic Evacuation Plan Model for Aboagye Menyah Complex at KNUST, Apr 2023 – Aug 2023
Department of Mathematics, KNUST Project
Supervisor: Prof. Peter Amoako-Yirenkyi

- Represented the complex as a network graph and developed a dynamic shortest-path model that takes the hazard's position on the network as input, recomputing evacuation routes as that position changes.
- Simulated evacuation under a range of hazard scenarios and specified a protocol for incorporating the model into a building's emergency and security system.

Teaching

Python Tutor Oct 2025 – Nov 2025
Ghana Computational Biosciences Network Kumasi

- Designed and delivered a two-month introductory Python programming bootcamp for 20–30 members of the network, writing all course materials and exercises.
- Assigned weekly exercises following each session and supervised a final project, through which participants applied the material to their own work.

Teaching Assistant Jan 2024 – Dec 2024
Department of Mathematics (KNUST) Kumasi

- Tutored undergraduate courses in differential equations and numerical analysis across a full academic year, working through problem sets with students and covering material from lectures.
- Instructed and graded laboratory sections on parallel programming with OpenMP and MPI in C, supporting students through the practical exercises and assessing their submitted work.

Professional

Part-time Database Developer Sep 2024 – Dec 2024
Adepts Ghana Kumasi

- Designed and built relational database schemas as part of the database team on two client systems: a hospital management system and a complaints management system for an insurance company.
- Maintained the databases supporting these systems in production, handling schema changes and data issues as they arose over the course of the engagement.

Publications and Presentations

A Communication-Avoiding Parallel Spectral Clustering Pipeline *In preparation*
P. Asumadu, E. S. Menkah, P. Amoako-Yirenkyi, R. Owusu
Manuscript — In preparation for submission

A Communication-Avoiding Parallel Spectral Clustering Pipeline Mar 2026
MPhil proposal seminar — NIMS-KNUST, Kumasi, Ghana

Real-Time Evacuation Planning Sep 2025
Poster — KNUST Scientific Conference, Kumasi, Ghana

Task Allocation Made Easy Dec 2024
P. Asumadu, O. Inusah
Poster — DS-Change Workshop: Uniting Academia & Industry, KNUST, Ghana

Awards and Scholarships

SCIM Scholarship 2025 – 2026
Center for Scientific & Technical Computing NIMS-KNUST

Skills

Programming: Python, C/C++, R, SQL

HPC: MPI, OpenMP, CUDA, OpenPBS, SLEPc, SciPy, Make, CMake

ML: Scikit-learn, NumPy, Pandas, Altair, Seaborn

Others: Git / GitHub, MATLAB, LaTeX, Shell scripting (Bash)

Workshops and Conferences

Indaba X: Ghana Data Science Summit, Ghana Oct 2024
Deep Learning Indaba, Accra, Ghana Sep 2023
High Performance Computing for Sustainable Development, ICTP, *Virtual* Apr 2023